

SHERBROOKE AP, QC, Canada

WMO: 716100

Lat: 45.4403N

Lon: 71.6918W

Elev: 241

StdP: 98.46

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.1	(c) -23.9	(d) -32.2	(e) 0.2	(f) -26.5	(g) -29.0	(h) 0.3	(i) -23.6	(j) 11.6	(k) -4.1	(l) 10.2	(m) -7.6	(n) 2.2	(o) 270	(p) 0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	29.0	21.4	27.5	20.4	26.1	19.6	22.7	26.9	21.7	25.7	20.7	24.6	4.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 21.3	(b) 16.4	(c) 25.2	(d) 20.3	(e) 15.4	(f) 24.0	(g) 19.3	(h) 14.5	(i) 22.9	(j) 68.3	(k) 26.8	(l) 64.4	(m) 25.8	(n) 60.9	(o) 24.6	(p) 27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.3	8.1	7.1	-32.0	31.6	3.9	0.9	-34.8	32.2	-37.1	32.8	-39.3	33.3	-42.1	34.0
			-32.2	24.5	3.9	1.1	-34.9	25.3	-37.2	26.0	-39.4	26.6	-42.2	27.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-10.3	-8.8	-3.2	4.3	11.7	16.4	19.0	17.9	13.9	7.4	0.8	-5.8
	DBStd	11.47	7.81	6.95	6.45	4.83	4.32	3.67	2.89	3.06	4.09	4.63	5.53	6.47
	HDD10.0	2666	628	525	411	181	31	1	0	0	11	109	278	492
	HDD18.3	4849	886	758	668	420	210	78	26	45	143	340	525	749
	CDD10.0	974	0	0	1	11	83	195	280	246	127	28	3	0
	CDD18.3	117	0	0	0	0	4	22	48	33	9	1	0	0
	CDH23.3	1137	0	0	0	8	82	243	415	288	97	4	0	0
	CDH26.7	216	0	0	0	1	14	53	84	46	17	0	0	0
Wind	WSAvg	2.9	3.2	3.3	3.4	3.4	3.0	2.5	2.3	2.2	2.3	2.8	3.0	3.0
Precipitation	PrecAvg	1162	79	64	82	78	97	104	120	133	101	92	108	103
	PrecMax	1555	160	117	134	195	188	181	227	208	177	175	217	194
	PrecMin	930	22	20	18	24	45	36	47	59	36	29	62	47
	PrecStd	169	33	25	30	33	41	39	50	41	38	37	31	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.3	10.1	16.4	24.0	28.8	30.5	31.0	29.9	29.0	23.8	18.5	10.5
		MCWB	9.5	6.2	10.3	14.4	18.9	22.3	23.4	22.3	21.8	16.6	14.2	8.4
	2%	DB	6.1	6.1	11.0	19.9	25.7	28.1	28.9	27.9	26.0	20.6	14.7	7.5
		MCWB	4.9	3.5	6.4	12.3	16.3	20.7	21.9	20.9	19.9	15.0	11.0	6.2
	5%	DB	3.0	3.5	8.0	16.1	23.0	26.3	27.2	26.4	23.8	18.0	11.7	4.9
		MCWB	1.9	1.5	4.3	9.8	15.6	19.4	20.7	20.1	18.5	13.8	8.6	4.1
	10%	DB	0.7	1.5	5.6	12.9	20.4	24.1	25.7	24.8	21.6	15.3	8.9	2.5
		MCWB	-0.3	0.0	2.9	7.7	14.0	18.3	19.9	19.4	17.4	12.0	6.8	1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	7.3	12.2	16.6	20.2	23.6	24.4	23.3	22.5	18.5	14.9	8.7
		MCDB	11.0	9.3	16.5	22.4	26.0	28.7	29.5	27.9	27.4	22.2	17.1	10.0
	2%	WB	5.1	4.1	7.4	13.3	18.3	21.9	23.0	22.0	20.9	16.2	11.9	6.4
		MCDB	5.9	5.5	9.8	18.2	23.1	26.4	26.7	26.0	24.6	19.2	14.2	7.2
	5%	WB	1.9	1.7	4.9	10.6	16.6	20.5	21.9	21.0	19.4	14.2	9.3	4.0
		MCDB	3.0	3.1	7.5	14.5	21.2	24.7	25.7	24.9	22.7	17.4	11.1	5.0
	10%	WB	-0.2	0.1	3.1	8.4	14.8	19.2	20.8	20.1	17.9	12.3	6.9	1.5
		MCDB	0.7	1.5	5.4	12.5	18.9	22.9	24.5	23.6	20.8	14.9	8.9	2.3
Mean Daily Temperature Range	5% DB	MDBR	10.4	11.1	10.6	10.8	12.4	11.9	11.6	11.9	12.2	10.4	8.5	8.5
		MCDBR	11.1	11.4	13.9	17.5	16.8	15.1	13.5	14.0	14.8	15.1	12.9	9.3
		MCWBR	10.5	9.4	9.7	10.6	8.8	8.3	7.1	7.7	8.9	9.7	9.9	8.8
	5% WB	MCDBR	11.0	10.6	11.9	15.1	13.6	12.7	11.6	11.9	12.8	13.3	11.4	8.9
		MCWBR	10.5	9.2	9.1	10.0	8.0	7.7	6.7	7.0	8.2	9.4	9.6	8.7
Clear-Sky Solar Irradiance	taub		0.264	0.282	0.311	0.345	0.369	0.410	0.416	0.396	0.364	0.326	0.302	0.274
	taud		2.371	2.287	2.337	2.390	2.387	2.267	2.257	2.317	2.405	2.510	2.496	2.392
	Ebn at Noon		857	902	921	915	899	859	848	853	853	845	806	800
	Edh at Noon		73	97	107	112	116	132	132	119	100	79	66	65
All-Sky Solar Radiation	RadAvg		1.39	2.28	3.49	4.35	4.90	5.43	5.42	4.82	3.90	2.25	1.37	0.97
	RadStd		0.19	0.28	0.26	0.48	0.55	0.41	0.47	0.29	0.33	0.22	0.18	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+1.62	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page